Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How load does it take to download the file?

Reasoning + Answer:

ÐµÐ¶Ðµ Ð½ÐµÐ¾Ð±ÑħÑĸÐ´Ð½Ð¾ Ð¾Ð±ÑĩÐ¸ÑģÐ»Ð¸ÑĤÐ¸, ÑģÐºÑĸÐ»ÑĮÐºÐ¸ ÑĩÐ°ÑģÑĥ Ð·Ð½Ð°Ð´Ð¾Ð±Ð¸ÑĤÑĮÑģÑı ÐļÐ°ÑĢÐ»Ñĸ, ÑīÐ¾Ð± Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶Ð¸ÑĤÐ¸ 200 ÐĵÐij ÑĦÐ°Ð¹Ð», Ð²ÑĢÐ°ÑħÐ¾Ð²ÑĥÑİÑĩÐ¸, ÑīÐ¾ Ð½Ð° 40% Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı Ð²ÑĸÐ´Ð±ÑĥÐ²Ð°ÑĶÑĤÑĮÑģÑı Ð¿ÐµÑĢÐµÐ·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı Ð½Ð° 20 ÑħÐ²Ð¸Ð»Ð¸Ð½, Ð° Ð¿Ð¾ÑĤÑĸÐ¼ Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı Ð¿Ð¾ÑĩÐ¸Ð½Ð°ÑĶÑĤÑĮÑģÑı Ð· Ð¿Ð¾ÑĩÐ°ÑĤÐºÑĥ.

ÐŁÑĸÑģÐ»Ñı Ð¿ÐµÑĢÐµÐ·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı Ð²Ð¾Ð½Ð° Ð¿Ð¾ÑĩÐ¸Ð½Ð°ÑĶ Ð·Ð½Ð¾Ð²Ñĥ Ð· Ð¿Ð¾ÑĩÐ°ÑĤÐºÑĥ, ÑĤÐ¾Ð¼Ñĥ Ð²ÑģÑı Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı Ð²ÑĸÐ´Ð±ÑĥÐ²Ð°ÑĶÑĤÑĮÑģÑı Ð´Ð²ÑĸÑĩÑĸ:

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1. ÐĿÐ¾ÑĢÐ¼Ð°Ð»ÑĮÐ½Ð° ÑĪÐ²Ð¸Ð´ÐºÑĸÑģÑĤÑĮ Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı: 2 ÐĵÐij/ÑħÐ²

ÐłÐ¾Ð·Ð¼ÑĸÑĢ ÑĦÐ°Ð¹Ð»Ñĥ: 200 ÐĵÐij

40% Ð²ÑĸÐ´ 200 ÐĵÐij = 0.4 ÃĹ 200 = 80 ÐĵÐij

ÐłÑĤÐ¶Ðµ, Ð´Ð¾ Ð¿ÐµÑĢÐµÐ·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÑĥÑĶÑĤÑĮÑģÑı 80 ÐĵÐij.

Ð§Ð°Ñģ, ÑīÐ¾ Ð²Ð¸ÑĤÑĢÐ°ÑĩÐ°ÑĶÑĤÑĮÑģÑı Ð´Ð¾ Ð¿ÐµÑĢÐµÐ·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı:

= 80 ÐĵÐij / 2 ÐĵÐij/ÑħÐ² = 40 ÑħÐ²

ÐŁÐ¾ÑĤÑĸÐ¼ **20 ÑħÐ²Ð¸Ð»Ð¸Ð½ Ð¿ÐµÑĢÐµÐ·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı (Ð½Ð° Ð¿ÐµÑĢÐµÐ·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı)**

ÐŁÑĸÑģÐ»Ñı ÑĨÑĮÐ¾Ð³Ð¾ Ð²Ð¾Ð½Ð° Ð¿Ð¾ÑĩÐ¸Ð½Ð°ÑĶ Ð·Ð½Ð¾Ð²Ñĥ Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÑĥÐ²Ð°ÑĤÐ¸ Ð²ÑĸÐ´ Ð¿Ð¾ÑĩÐ°ÑĤÐºÑĥ. ÐĵÑĸÐ´Ð¾Ð¼Ð¾, ÑīÐ¾ Ð²Ð¾Ð½Ð° Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÑĥÑĶÑĤÑĮÑģÑı Ð·Ð½Ð¾Ð²Ñĥ Ð²ÑĸÐ´ Ð¿Ð¾ÑĩÐ°ÑĤÐºÑĥ, ÑĤÐ¾Ð¼Ñĥ Ð¿Ð¾Ð²Ð½Ð° Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı âĢĶ Ð·Ð½Ð¾Ð²Ñĥ 200 ÐĵÐij.

Ð§Ð°Ñģ Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı Ð¿ÑĸÑģÐ»Ñı Ð¿ÐµÑĢÐµÐ·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı: 200 / 2 = 100 ÑħÐ²

ÐĹÐ°Ð³Ð°Ð»ÑĮÐ½Ð¸Ð¹ ÑĩÐ°Ñģ = ÑĩÐ°Ñģ Ð´Ð¾ Ð¿ÐµÑĢÐµÐ·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı + ÑĩÐ°Ñģ Ð¿ÐµÑĢÐµÐ·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı + ÑĩÐ°Ñģ Ð²ÑĸÐ´Ð½Ð¾Ð²Ð»ÐµÐ½Ð½Ñı Ð·Ð°Ð²Ð°Ð½ÑĤÐ°Ð¶ÐµÐ½Ð½Ñı

= 40 + 20 + 100 = **160 ÑħÐ²Ð¸Ð»Ð¸Ð½**

<answer>160</answer>